ISOMETRIC

Bio-oil Storage in Permeable Reservoirs

Storage Module

v1.1

This module details durability and monitoring requirements for bio-oil storage in permeable reservoirs.

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1.0

Introduction

Durability refers to the length of time for which CO_2 is removed from the Earth's atmosphere and cannot contribute to further climate change. This module details durability and monitoring requirements for bio-oil storage in permeable reservoirs.

On injection, bio-oil is expected to be a dark, viscous liquid of typically between pH 2-3 (but up to 6), consisting of oxygenated hydrocarbon compounds[1]. The durability of bio-oil stored in geologic formations depends on the characteristics of the bio-oil, the geologic storage formation and the interactions between the two. To ensure sufficient durability, bio-oil characteristics and conditions of storage must be well defined, modeled, and monitored.

This module is applicable for bio-oil injection into saline aquifers and depleted hydrocarbon fields (clastic and carbonate reservoirs) that have been approved by the EPA or the State to which EPA has delegated permit authority. This module is not applicable for bio-oil that has any bio-char mixed with it.

Bio-oil is expected to be denser than the surrounding subsurface brine, oil and gas, meaning it is expected to sink after injection. This eliminates the risk of vertical migration due to injectate buoyancy after injection, which is a risk for supercritical CO_2 storage in the subsurface[2 3]. The storage of bio-oil in geologic reservoirs is has not been widely studied and documented as of December 2023. Research by Charm Industrial, has published aging effects with temperature and time with bench scale experiments[2]. At a viscosity of 6000 cP, bio-oils become difficult to pour and stir, and they are expected to reach this between 2 and 15 years at reservoir

temperatures between 60°C and 35°C respectively[2] [4]. After polymerization into a solid[5], bio-oil is immobile within the subsurface, which when coupled with capping and closure of subsurface reservoirs as per UIC permitting requirements. The carbon stored within the bio-oil is expected to be removed from the atmosphere on geological timescales.

Potential risks to the expected durability of bio-oil are as follows:

- Bio-oil may not undergo polymerization post-injection, and remains as a liquid which could migrate out of the intended storage reservoir. In any case, until bio-oil solidifies (as noted above, this could take between 2-15 years), risk of migration out of the intended storage reservoir is a possibility.
    - Bio-oils as a category cover a wide variety of characteristics, with significant variation across bio-oils produced based on the feedstock, biomass conversion process, operating conditions, and processing completed[6]. Specific characteristics of bio-oils such as total acid number (TAN), pH, and oxygen content, can impact stability of bio-oil as well as its tendency to polymerize (solidify). Although instability due to polymerization is an unfavorable effect when producing bio-oils for fuels production, for durable storage it is beneficial[7].
- Bio-oil may be converted to bio-gases in the subsurface reservoir such as CH_4, CO_2 and short chain hydrocarbons.
    - Bio-oil could biodegrade to form CO_2 and short-chain hydrocarbons, subsequently leading to microbial methanogenesis and release of CH_4 gas. The composition of bio-oil means this should be unlikely and if it does occur should only do so in small

volumes[8] [2], because of (i) a lack of wettability and (ii) nutrient limitation in the subsurface. Any signs of bio-gas formation before and/or after bio-oil polymerization (solidification) should be monitored as part of the post-injection monitoring plan (see Section 3.2). Any gases produced shall be considered removed because of the presence of one or more confining layers that prevent the vertical migration of buoyant phases (such as biogas). Any releases of gases from the geologic reservoir must be accounted for as detailed in Section 3.3.

- Bio-oil may react with surrounding reservoir rocks:
  - The low pH of bio-oil could induce reactions with surrounding reservoir rocks, which could decrease durability by providing conduits for migration out of the storage reservoir via faulting and/or fracturing. Research by Charm Industrial has shown that 7 and 21-day exposures of sandstone and dolomite to bio-oil at bench scale demonstrated < 5% and < 3% weight loss respectively, and are proposed as upper bounds of reactivity due to the poor cementation of the samples[2].

This section outlines requirements for evaluating bio-oil injection and storage, with a focus on site characterization, construction and monitoring. The post-injection monitoring plan detailed in Section 3.2 acts to address and mitigate these potential risks to durability. Section 3.3 addresses accounting for any emissions associated with these risks.

Monitoring of the injection site shall be completed to ensure that any injected bio-oil remains stored within the confines of the geologic reservoir and does not migrate outside of the reservoir limits, nor result in decay of bio-oil and subsequent re-emission as CH_4, CO_2 or other volatiles. The injection site

shall be monitored in accordance with the U.S. EPA Underground Injection Control (UIC) permitting requirements as specified in the operating permit for the injection site issued by the EPA or the State to which EPA has delegated permit authority[9].

The subsurface monitoring approach developed and implemented by the project proponent shall address, via the permitting process and permit compliance, or by additional efforts and documentation:

- Geologic Reservoir and Site Characterization: the proposed storage site must have been properly characterized to demonstrate site suitability for storage and containment of bio-oil or other biomass or organic materials. See Section 2.2 for further details.
- Injection Site Construction and Performance: the proposed storage site and injection system must be properly designed, including design and specification of wellbore and well materials to ensure proper long term operation of the well when injecting bio-oil and protection of the lowermost underground source of drinking water (USDW).
- Injection System Operation & Monitoring: the project proponent must specify operating conditions and monitoring systems and approaches, such as allowable wellhead pressures, gas detection, and other systems to ensure that injected bio-oil remains in the geologic formation, the formation is not negatively impacted by operations, automatic safety precautions are in place to minimize potential for exceeding allowable operating conditions, and conditions can be monitored for compliance or deviation from requirements. Reporting of operations will be in accordance with the UIC governing body.

- Closure and Post-Closure Requirements: Requirements for proper closure of the storage reservoir and injection facility, as well as post closure requirements and post-injection monitoring to ensure bio-oil remains sequestered durably in the storage reservoir, the site is properly monitored, and any non-compliance is addressed with corrective actions.

Specifically, the following requirements must be met to ensure durable storage of bio-oil in the geologic reservoir.

2.0

Permitting and Site Characterization

2.1

Permitting

The injection site must have a current Class I or V UIC[10] well permit issued by the responsible authority for the location of the injection facility and reservoir. The permit must specifically identify bio-oil or an equivalent type of injectant, as acceptable injectants under the permit[11].

2.2

Site Characterization and Feasibility Requirements

The site should be well characterized in accordance with the permit application and approval requirements under the UIC regulations. Site characterizations must include evaluation of reservoir chemistry and conditions where required to ensure compatibility of bio-oil with the storage reservoir. The permit shall define the Area of Review for the site in accordance with the requirements for the specific well class, formation, and local characteristics[12].

As part of the UIC permit application, the project proponent must demonstrate that the geologic system:

- includes a sequestration zone of sufficient volume, porosity, permeability, and injectivity to receive the total anticipated volume of the bio-oil stream

- includes a confining system composed of a layered interval of low permeability rocks that will prevent vertical migration of bio-oil, brine or biogas above the storage complex, towards the surface and atmosphere and/or USDWs
- includes a confining system free of transmissive faults and fractures and of sufficient extent and thickness to contain the injected bio-oil stream, displaced formation fluids and any biogas generation, and allow injection at proposed maximum pressures and volumes without initiating or propagating fractures in the confining zone(s)
- will not be impacted by, or induce as a result of the injection process, seismicity at levels that may inhibit the durability of bio-oil storage due to changes in the formation structure. If this seismic risk exists, the project proponent will establish criteria within the UIC permit that require relevant seismic monitoring or preventive limitations on injection

In addition, the project proponent must also characterize the following to assess the risk of leakage, develop the operation conditions for injection and monitoring plans, model the injectate behavior and for comparison to future measurements:

PARAMETER	PURPOSE
Reservoir lithology and mineralogy	Allowing for predictions on whether and how bio-oil may interact with the targeted formation
Porosity and permeability of sequestration zone strata	To demonstrate the capacity of the target formation to receive and safely store bio-oil
Confirmation of low permeability and structural integrity of confining layer/cap rock	To demonstrate that if any buoyant fluids or gasses form, they will be trapped and unable to migrate upwards out of the reservoir

PARAMETER	PURPOSE
Temperature, pH, conductivity and fluid saturation of storage reservoir formation fluid/brine	For density and reactivity calculations. Potential interaction of the injectate under these conditions with the storage complex may impact whether any potential products (e.g., biogas) are produced as well as injectate mobility and stability.
Dissolved gas, including of dissolved inorganic carbon (DIC), composition in formation fluids and composition of any tracers being used (e.g., $\delta^{13}C$ signature and/or major and minor ion).	Determine the source of any produced biogas and extent of secondary trapping mechanisms or reactions (e.g., dissolution, methanogenesis).
Formation fluid stable oxygen $\delta^{18}O$ and hydrogen δD isotope signature, where required in the permit	For determining the extent of bio-oil rock interaction on injection/during monitoring.
$\delta^{13}C$ of the compounds of the bio-oil, where applicable	For determining the source of any produced biogas and extent of reactions (e.g. methanogenesis) as a result of injection.
Quantity of hydrocarbons present in the reservoir prior to bio-oil injection	For determining the source of any produced biogas and extent of reactions (e.g. methanogenesis) as a result of injection.

In addition, due to the unique characteristics of the bio-oil, during permitting and approval of bio-oil injection into the formation under the UIC permit, site geology and geochemistry, potential interaction with the bio-oil and in-situ fluids, injectant mobility, and bio-oil stability may be required. The UIC permit may also require a “dissipation interval” with hydrogeologic properties sufficient to attenuate pressure created by bio-oil or formation fluid migration to below the storage complex to limit downward overpressure propagation.

The project proponent must demonstrate and justify that the bio-oil and injection process result in long term stability, limited lateral migration, and limited degradation such that bio-oil injection does not impact the designated lowermost USDW or above-surface environmental conditions. Justification may include reservoir simulation work if required by the UIC permit, which considers site and injectant characteristics; alternatively, academic studies and peer-reviewed literature representative of the site and injectant

characteristics, mobility studies, bio-oil aging studies[13], or other predictive data and studies completed in conjunction with performance monitoring of the formation, such as pressure front monitoring, to ensure fracturing and resulting mobility are not occurring. Specific laboratory core analysis experiments with relevant cores should be conducted to confirm suitability for bio-oil sequestration operations, including quantification of bio-oil reactivity with the core. The laboratory experiments may also include quantification of the rate at which bio-oil polymerizes (solidifies), and exploration of bio-oil flow. A relevant core could be a representative rock sample from a sister reservoir, or equivalent, or core directly sampled from the project site.

Site characterizations and analytical modeling shall be reviewed every ten years as part of the UIC permit renewal application minimum, at the UIC Program Director's request, or when monitoring and operational conditions warrant, as indicated by a significant change in site conditions or injectant characteristics, based on monitoring data. The review shall include a comparison of pre-injection project assumptions to actual measured conditions including plume size, extent, and migration, where possible, and specific operating conditions observed during injection. Estimates revised with any acquired monitoring data should demonstrate that the planned injection volume will remain within the storage complex until the end of the post-injection monitoring period.

2.3
Well Construction Requirements

The project proponent must ensure that the injection well is constructed in compliance with the EPA UIC permit and

documentation and records of well construction are maintained and available for review.
At a minimum, the project proponent must ensure that all injection, observation or monitoring, legacy offset and production wells contained within the delineated AOR have been evaluated and wells which pose a risk to durability plugged prior to injection in order to:

- Prevent the movement of fluids into or between any unauthorized zones
- Prevent the movement of fluid into USDW
- Permit the use of appropriate testing devices and workover tools
- Permit continuous monitoring of the injection well pressure in the annulus space between the injection tubing and long string casing

Casing, cement, tubing, packer, wellhead, valves, piping, or other materials used in the construction of each well associated with the project must have sufficient structural strength and be designed for the life of the project. All surface casing will be set below the lowermost USDW and cemented to the surface. All well materials must be compatible with fluids with which the materials may be expected to come into contact, including bio-oil and formation fluids (e.g., corrosion-resistant well casings) and must meet or exceed standards developed for such materials by API, ASTM International, or comparable standards. The casing and cementing program must be designed to prevent the movement of fluids out of the sequestration zone and above the storage complex.

If pre-existing wells are being used for injection or monitoring, special considerations are necessary to ensure the integrity of the well and to prevent fluid migration along the borehole.

These must be agreed and checked by the regulating body to determine construction and safety is consistent with well construction requirements. All checks and modifications must be recorded and all records kept. This must include the following:

- determining the integrity of the cement and casing.
- conducting any necessary action to repair defects.
- determining whether the existing well materials are adequate for the new function of the well.
- determining the diameter of the hole, any deviations from vertical, and any significant curvature or bends in the well should be compared with the size of the proposed monitoring equipment.
- existing well materials should be checked to ensure that they are compatible with fluids with which they will come into contact, such as carbon dioxide and carbon dioxide-rich brines if they are completed in the injection zone.
- any flaws in the casing or cement will need to be repaired.

3.0

Monitoring

3.1

Injection and Operational Monitoring Requirements

The project proponent will ensure that the injection facility complies with the well permit, including the development and implementation of the well operating plan as required by the permit. At a minimum, the well operating plan shall consider the following:

3.1.1

Injection and Injectant Operation & Monitoring

- Maximum allowable surface injection pressure (MASIP) at the injection wellhead that is allowed during injection operations to prevent fracturing of the formation, set according to the UIC permit. Injection operation pressures shall reflect local regulatory agency requirements for formation fracture pressure as a precaution to ensure that the geologic formation will not be fractured.
    - Installation and use of continuous recording devices to monitor injection pressure and the pressure on the annulus between the tubing and the long string casing
    - Monitoring and documentation of injection pressure must be performed and records maintained for review
- Maximum bio-oil injection rate to monitor volumes injected, prevent induced seismicity or return of injectant
    - Installation and use of continuous recording devices to monitor injection rate and volume

    - Monitoring and documentation of injection rate must be performed and records maintained for review
- Limitations on composition of the injected fluid, including, but not limited to pH, density, temperature or other parameters, if relevant, to ensure injectant does not negatively impact the formation via inducing dissolution, reaction, or other degradation pathways, resulting in increased potential for bio-oil and fluid migration. Density in particular must be greater than that of the formation water.
    - Records of laboratory analyses and relevant permit limitations to demonstrate compliance must be maintained in accordance with the well permit and available for review.
- Analysis of the bio-oil stream with sufficient frequency to yield data representative of its chemical and physical characteristics, including analysis of the following for each injection batch using industry standard or indicated methods and quality and properly calibrated equipment:
    - pH
    - density
    - viscosity
    - TAN
    - carbon © content (for further details see Section 7.3.3.1 of the main protocol)
    - δ^{13}C of the bio-oil compounds (where applicable)

  Additional measurements of the bio-oil stream may include, oxygen content or a geochemical characterization/fingerprint of bio-oil (for example the major and minor ions).
- Wells must have gas detectors (or equivalent sensors/imaging) with alarms and injection shut-off systems (e.g., automatic shut-off or procedures in place for

manual shut off of injection/operation), including for gaseous release (CO_2, hydrocarbons or other GHGs) and injection pump shutoff when maximum pressure is reached or maximum flow rate is exceeded. If activated the operator must immediately investigate and identify as expeditiously as possible (or in accordance with permit requirements) the cause of the alarm or shutoff, and report the instance to the validation and verification body (VVB).

3.1.2
System Integrity Monitoring

- Corrosion monitoring of well materials, upon well completion and performed and reported every 6 months thereafter, for loss of mass, thickness, cracking, pitting, and other signs of corrosion, to ensure that well components meet the minimum standards for material strength and performance set by API, ASTM International, or equivalent. If injection frequency is less than twice per week in that timeframe, then corrosion monitoring requirements are not mandatory, but are recommended. Alternatively, corrosion performance of well materials may be evaluated using the Corrosivity Screening of Pyrolysis Bio-oils by Short Term Alloy Exposures developed by NREL[14]
- A demonstration of external mechanical integrity on a cadence specified by the UIC permit annually until the injection well is plugged
- A pressure fall-off test, annually

3.1.3

Migration and Storage Reversal Monitoring

- As applicable based on specific site conditions, formation type, and permit class, monitoring to ensure bio-oil migration beyond the AOR has not occurred. Such approaches may include periodic monitoring of pressure and/or water chemistry in overlying formations above the storage complex, noting changes versus baseline conditions that may indicate bio-oil related migration.
- As applicable based on specific site conditions, formation type, permit class, and as required by the UIC permit, analysis of groundwater for bio-oil related parameters on an annual basis, to include at a minimum, analysis for bio-oil fingerprint. Results must be compared to baseline values obtained prior to bio-oil injection[15].
- Monitoring of the composition of any gas recovered from well(s) or representative sampling locations where detection of gases from the reservoir may be completed (such as a recovered brine sealed holding tank). Gas composition monitoring shall include CO_2, CH_4 and VOCs emissions from the storage reservoir via CO_2, CH_4 and VOCs gas monitors with a resolution of at least 0.01 vol% or lab analysis, if sampled. Results must be compared to baseline values obtained prior to bio-oil injection and any differences must be assessed by the Project Proponent to see if they are attributable and material to the project. Sampling frequency should be aligned with other well and reservoir testing (e.g., corrosion monitoring) every 6 months.
- As applicable based on specific site conditions, formation type, and permit class, monitoring of wellbores within the AOR that intersect the storage complex at depth, annually while operating and every 5 years post-closure.

Monitoring should include direct observation of the wells, if possible, and surface air monitoring around the wellbore. Monitoring should focus on pressure measurements and identifying gaseous degradation products (CO_2, CH_4, VOCs) in the vicinity of the wellbore that may indicate bio-oil degradation and/or migration. Results must be compared to baseline values obtained prior to bio-oil injection.

- A seismic monitoring program may be suggested at the discretion of the UIC director in areas of increased seismic risk where seismicity may have an impact on the formation and the durability of the bio-oil being stored.The area within the AOR of the injection facility and the area of the storage complex must be monitored for magnitude 2.7[12] or greater seismic events to determine the presence or absence of any induced micro-seismic activity associated with the wells and near any discontinuities, faults, or fractures in the subsurface.

For all injection and injectant (bio-oil) monitoring and analyses, sufficient samples must be analyzed to determine that the composition of the bio-oil is within specified parameters in the UIC permit, where required. Each individual batch of bio-oil that is injected should be analyzed and characterized to ensure variations in pH, density, viscosity, TAN and δ^{13}C of the bio-oil compounds (where applicable) from batch to batch are accounted for. Samples should be from a well mixed and representative container of the bio-oil.

Any major consistent deviations could result in an AOR re-evaluation, which may be performed based upon injection operations at the UIC Program Director's discretion/request.

If any CO_2 leaks are detected from the targeted reservoir or there are significant irregularities from the used model(s), the Project Proponent/operators must undertake corrective measures as set out in their monitoring plan submitted and approved by the UIC Program Director. For a loss of conformance with models, the Project Proponent must halt injection whilst they identify the cause of this loss, and then revise the monitoring plan to account for this. If the caprock is found to have lost integrity, the Project Proponent must halt injection whilst they conduct an assessment to determine whether there is any leakage and whether the loss of containment and/or well mechanical integrity can be repaired prior to injection beginning again. If loss of caprock integrity is found to be the cause, injection must cease. The amount of CO_2 lost must also be quantified and subtracted from the overall total of CO_2 stored.

Re-evaluations of the bio-oil fluid plume extent must also be implemented when warranted based on observational or quantitative changes of the monitoring parameters of the storage reservoir, including but not limited to:

- observed migration of the bio-oil fluid plume is unexpected and suggests potential movement of bio-oil outside the intended formation
- bio-oil migration into a zone above the storage complex
- actual bio-oil fluid plume or elevated pressure extend beyond analytical model expectation because any of the following has occurred:
    - any seismic activity in the area of magnitude 2.7[16] or greater within the AOR;
    - new site characterization data change the model inputs to such an extent that the predicted bio-oil

fluid and/or pressure plume extends vertically or horizontally beyond what was originally predicted.

3.2 Post-injection Monitoring Plan

The aim of this post-injection monitoring plan is to put in place scientific and/or operational monitoring practices which go beyond requirements of the UIC injection permits specified under the UIC well classes allowed under this protocol, in order to prove beyond reasonable doubt that bio-oil storage will be durable for the expected 100,000-1,000,000 year timescales. As outlined in Section 1, bio-oil injection is a nascent area of research, and so addressing potential risks to durability is important for ensuring robust and diligent CO_2 removals. Direct imaging of bio-oil using seismic surveys will be challenging due to the density contrast between bio-oil and the surrounding formation fluid/brine being minimal[2]. Therefore, post-injection monitoring must focus on using both injection and monitoring wells, if applicable (the plans for which are addressed by the UIC permit) to measure presence and characteristics of bio-oil directly. Please note, the requirements in this section should be followed prior to closure of the injection well (see Section 3.5).

The Project Proponent must follow any post-injection requirements of the UIC permit for the specified project, in addition to the following:

Before polymerization (solidification) of bio-oil:

- Density contrast is the key mechanism that provides long-term durability of bio-oil storage. The density of bio-oil samples must be measured prior to each injection and

compared to formation fluid values to ensure that bio-oil will sink upon injection. These measurements should be repeated before each new injection of bio-oil. Formation fluid samples will be acquired prior to bio-oil injection and compared to historical production data from the reservoir or nearby representative analogues. Density contrast is demonstrated by:

- Calculation according to the density contrast equation below:
  - $\rho_{bio-oil} > \rho_{formationfluid}$ where ρ is density at reservoir conditions calculated
  - $\rho = m/V$ where m is mass and V is volume, calculated according to reservoir-specific conditions, such as pressure, temperature and salinity (and any other reservoir conditions relevant to a density calculation).
  - Bio-oil density will be calculated according to reservoir conditions (as above) using the measured surface density (measured following ASTM D 4052-22 or ASTM D7042-21a) before commencing injection. Formation fluid density will be calculated or measured using samples of the formation fluid collected prior to injection.

- Bio-gas monitoring:
  - Intermittent gas sampling for emissions of CO_2, CH_4 and VOCs is required, when gas is detected, during injection of bio-oil until closure of the well. If any emissions outside of the normal average baseline range for any of these gases occur, then further monitoring of gases should take place. Further monitoring entails measurements of:
    - concentrations (e.g., CO_2, CH_4 VOCs), usually measured by GC

    - stable isotope compositions of CO_2, CH_4 and VOCs (δ^{13}C of CO_2, CH_4 , VOCs and δD of CH_4), usually measured by an isotope ratio mass spectrometer (IRMS) or cavity ring down mass spectrometry, to determine the source of the bio-gas emissions. δ^{13}C must be measured against the standard reference material Vienna Peedee Belemnite (VPDB). δD must be measured against the standard reference material Vienna Standard Mean Ocean Water (VSMOW).
- Monitoring for interaction with surrounding reservoir rocks:
  - Intermittent gas sampling every 6 months of CO_2, CH_4 and VOCs, as above, with the same trigger conditions for further measurements of CO_2, CH_4 and VOCs.
  - Baseline formation fluid composition, to include temperature and major cations and anions, other characterization may include water isotope ratios (δ^{18}O and δD)
  - Formation fluid composition samples taken where possible during corrosion monitoring every 6 months (see Section 3.1.2), to include major cations and anions and temperature as well as if possible characterization of the water isotope ratios (δ^{18}O and δD)
- System Integrity monitoring should continue, with frequencies set based on information gathered during the operational phase
- The spread of the plume post-injection should be monitored to ensure that it is within laterally contained land and/or pore space ownership limits and vertically protected from the defined lowermost USDW using:
  - Monitoring via monitoring wells, where possible
  - Modeling (or alternative methods)

At a time period expected between 2-15 years, sampling and/or other monitoring data from the injection well and/or monitoring wells at the time intervals specified here may demonstrate that bio-oil has polymerized (solidified)[17] for example by obtaining a core from the reservoir. After the given time period, the following monitoring may be required under certain site conditions:

- Periodic biogas monitoring every 6 months (exactly as in Section 3.1.3)
- It is possible that before polymerization there will be separation of a carbon-rich aqueous phase from bio-oil. This is expected to be denser than formation fluids and will therefore remain unable to migrate upwards towards the surface[2] [18] [19]. Monitoring of the aqueous phase will be challenging due to it remaining deep in the reservoir and being indistinguishable from formation brines on geophysical logs. Any migration of fluids in an unexpected manner observed at the surface prior to closure of the site should be sampled and measured for (i) carbon content and density (as above), and any reversals in storage accounted for as outlined in Section 3.3.

3.3 Risk of Reversal

Based on the present understanding, projects applicable to this protocol are categorized as having a Low Risk Level of Reversal according to the Isometric Standard Risk Assessment Questionnaire. This is because there should be no reversals unless there is a loss of caprock or well integrity, and this technology does not yet have a documented history of reversals. However there is a risk of methane production within the reservoir. A 5% buffer pool will be set aside as a precaution

against CO_2 or CH_4 gas released. This reversal risk will be reassessed when new scientific research and understanding arises.

Reversals will be accounted for by projects and the Isometric Registry as detailed in Section 5.6 of the Isometric Standard.

3.3.1 Attribution of reversals

When a reversal is detected and quantified, there are multiple considerations that will be taken into account to attribute the reversal to whatever has been injected in the targeted reservoir.

1. If the Project Proponent was the only entity injecting into a given reservoir, the Project Proponent will take on 100% of the reversal.
2. If the Project Proponent was one of multiple entities injecting into that reservoir, the Project Proponent will be allocated a percentage of the reversed CO_2 proportional to the mass of injected material. For example:
    - A reservoir has a total of 200t of material emplaced at the time when the reversal is detected (this information should be provided by the Operator).
    - The Project Proponent has injected 50t of material in that reservoir.
    - The amount of reversed CO_2 has been quantified to be 10t.
    - The Project Proponent must compensate for 25% (50/200) of 10t CO_2 = 2.5t of CO_2.

In instances where reversals are determined to be a result of negligence by the Operator or Project Proponent, project

crediting may be ceased.

3.4

Calculation of $CO_2e_{Emissions}$

$CO_2e_{Emissions}$ is the total greenhouse gas emissions associated with a given Reporting Period, RP, or batch, n.

Equations and emissions calculation requirements for $CO_2e_{Emissions}$, including considerations for monitoring activities, are set out in the relevant protocol and are not repeated in this module.

3.5

Closure Requirements

The project proponent shall ensure that all UIC permit requirements associated with planning for, preceding with and monitoring of well or storage reservoir closure are adhered to and documented as required by the permit. A Site Closure Plan shall be prepared in accordance with the UIC permit requirements. Site closure must follow this plan and any relevant regulatory jurisdiction requirements for site decommissioning. This must include plugging of any wells within the AOR.

CO_2 storage agreements with pore space owners will ensure activity in the storage site is prohibited for perpetuity following bio-oil injection, ensuring that even if bio-oil does not solidify, it will not be subject to pressure disturbances (i.e., injection or production activities) in the sequestration reservoir.

The project proponent must monitor the site following injection completion to determine the three-dimensional extent of the bio-oil fluid plume and elevated pressure and demonstrate

that no bio-oil fluid migration out of the storage reservoir is occurring, as per the post-injection monitoring plan in Section 3.2 above.

After 15 years post-closure, unless otherwise specified within the relevant permit, an assessment shall be completed to demonstrate that the bio-oil plume has stabilized or is trending towards stabilization - eliminating the risk of migration or release of bio-oil or its degradation products from the storage formation to the atmosphere. The project proponent will actively explore emerging technologies for measuring plume stabilization. The plume stabilization assessment shall be conducted in one of the following ways:

- Utilize predictive modeling based on monitoring data collected during post-closure monitoring to demonstrate the stability of the bio-oil plume and lack of plume migration in the formation that would present risk to water sources in the Area of Review.
    - Modeling must be validated by comparison to historical monitoring data
    - Models must utilize site specific geochemistry and bio-oil characteristics from analyses required in Section 2.2, 3.1.3 & 3.2 of this module
    - Models must assess the potential plume extent after 50 years and demonstrate that the plume will not migrate beyond the AOR and will not impact drinking water sources nor cause other environmental harms.
- Where conditions of the formation and existing monitoring wells allow:
    - Utilize samples collected of bio-oil from the in-situ bio-oil plume to demonstrate lack of degradation products that may impact plume migration or

formation degradation, including, but not limited to analysis of VOCs and δ^{13}C of bio-oil compounds
    - Perform downhole indirect studies (such as sonar, seismic, etc.) of the bio-oil plume to demonstrate limited change since pre-closure studies
    - Utilize tracer studies to demonstrate lack of any vertical migration of tracers in the bio-oil plume to any areas outside of the authorized injection zone or outside of the predicted lateral extent within the AOR
- or new methods as outlined in subsequent protocol versions and as measurement and monitoring technologies advance.

If the plume stabilization can be demonstrated by the above methods, and is independently reviewed and certified by a registered Professional Geologist (i.e. Chartered Geologist or equivalent), the bio-oil plume will be considered stabilized and additional monitoring post-closure may be discontinued if allowed under the applicable UIC permit.

Where plume stabilization is not demonstrated, the project proponent is responsible for ongoing monitoring for a period of 50 years post closure, or until plume stabilization is demonstrated. The length of ongoing monitoring will be subject to change given subsequent reanalysis.

3.5.1 Record keeping

All records associated with the characterization, design, construction, injection operation, monitoring, and site closure shall be developed, submitted to proper authorities as required by the UIC permit.

All records shall be maintained for a minimum of 10 years after the injection. All closure and post-closure monitoring records shall be maintained by the project proponent for a minimum of 10 years after closure.

4.0

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5.0

Definitions and Acronyms

Bio-oil
A mixture of water, organic acids, aldehydes, ketones, sugars, phenols, and other organic compounds derived from the thermal breakdown of biomass. Thermal breakdown of biomass is achieved via thermochemical processes, such as pyrolysis, which heat biomass in low- or no-oxygen environments to high temperatures (~e.g. 350-650°C). Bio-oil is often also referred to as pyrolysis oil or bio-crude.

Buffer Pool
A common and recognized insurance mechanism among Registries allowing Credits to be set aside (in this case by Isometric) to compensate for Reversals which may occur in the future.

Durability
The amount of time carbon removed from the atmosphere by an intervention – for example, a CDR project – is expected to reside in a given Reservoir, taking into account both physical risks and socioeconomic constructs (such as contracts) to protect the Reservoir in question.

Feedstock
Raw material which is used for CO_2 Removal or GHG Reduction.

Geologic Formation
A body of similar rock type (e.g. color, grain size, mineral composition, texture) and a particular location in the stratigraphic column (vertical rock layers). Formations are large enough to be mappable on Earth's surface or traceable in the subsurface.

Model
A calculation, series of calculations or simulations that use input variables in order to generate values for variables of interest that are not directly measured.

Module
Independent components of Isometric Certified Protocols which are transferable between and applicable to different Protocols.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Project Proponent
The organization that develops and/or has overall legal ownership or control of a Removal or Reduction Project.

Protocol
A document that describes how to quantitatively assess the net amount of CO_2 removed by a process. To Isometric, a Protocol is specific to a Project Proponent's process and comprised of Modules representing the Carbon Fluxes involved in the CDR process. A Protocol measures the full carbon impact of a process against the Baseline of it not occurring.

Registry
A database that holds information on Verified Removals and Reductions based on Protocols. Registries Issue Credits, and track their ownership and Retirement.

Reservoir
A location where carbon is stored. This can be via physical barriers (such as geological formations) or through partitioning based on chemical or biological processes (such as mineralization or photosynthesis).

Reversal
The escape of CO_2 to the atmosphere after it has been stored, and after a Credit has been Issued. A Reversal is classified as avoidable if a Project Proponent has influence or control over it and it likely could have been averted through application of reasonable risk mitigation measures. Any other Reversals will be classified as unavoidable.

Standards (scientific)

Standard physical constants as well as standard values set forth by bodies such as the National Institute of Standards and Technology (NIST) or others.

Storage

Describes the addition of carbon dioxide removed from the atmosphere to a reservoir, which serves as its ultimate destination. This is also referred to as “sequestration”.

6.0

Appendix 1: Monitoring Plan Requirements

This appendix details how the Project Proponent must monitor, document and report all metrics identified within this Module to demonstrate the durability of CO_2 removal. Following this guidance will ensure the Project Proponent measures and confirms carbon dioxide removed and long-term storage compliance, and will enable quantification of the emissions removal resulting from the Project activity during the Project Crediting Period, prior to each Verification.

This methodology utilizes a comprehensive monitoring and documentation framework that captures the GHG impact in each stage of a Project. Monitoring and detailed accounting practices must be conducted throughout to ensure the continuous integrity of the carbon dioxide removals and crediting.

The Project Proponent must develop and apply a monitoring plan according to ISO 14064-2 principles of transparency and accuracy that allows the quantification and proof of GHG emissions removals.

PARAMETER	PARAMETER DESCRIPTION	REQUIRED
pH	pH of formation fluid	Always
TAN	Total Acid Number of Bio-oil	Always

PARAMETER	PARAMETER DESCRIPTION	REQUIRED
pH	pH of Bio-oil	Always
Density	Density of formation fluid	Always
Density	Density of Bio-oil	Always
Temperature	Temperature of formation fluid	Always
Salinity	Salinity of formation fluid	Always
Chloride content	Chloride concentration in formation fluid	Always
Formation fluid composition	Formation fluid composition - include major cation and anions	Always
Conductivity	Conductivity of formation fluid	Always
δ^{18} oxygen isotope signature	δ^{18} oxygen isotope signature of formation fluid	Always

PARAMETER	PARAMETER DESCRIPTION	REQUIRED
Oxygen content	Oxygen content of bio-oil	Always
δ^2 hydrogen isotope signature	δ^2 hydrogen isotope signature of formation fluid	Always
Concentration of δ^{13} Carbon signature of DIC, DOC, and carbon speciation	Concentration of δ^{13} Carbon signature of DIC, DOC, and carbon speciation in formation fluid	Always
δ^{13} Carbon signature	δ^{13} Carbon signature of the compounds of the bio-oil	Always
Carbon content	Carbon content of aqueous decomposition products if produced	Under certain conditions: if gas produced
Viscosity	Viscosity of bio-oil	Always
Hydrocarbon baseline	Estimate and characterization of any hydrocarbons present in the reservoir prior to bio-oil injection.	Under certain conditions: if a Class II well is used

PARAMETER	PARAMETER DESCRIPTION	REQUIRED
GC-MS	Analysis of bio-oil constituents via GC-MS	Always
GC-MS	Analysis of formation water for bio-oil constituents via GC-MS	Under certain conditions: if permit specifies groundwater quality monitoring
Porosity & permeability	Porosity & Permeability of sequestration zone strata	Always
Well volume	Sequestration zone of sufficient volume	Always
Injectivity	Sequestration zone of sufficient injectivity to receive the total anticipated volume of bio-oil	Always
Plume monitoring	Plume size, extent and migration	Always
Injection pressure	Surface injection pressure - shall reflect local regulatory agency requirements	Always
Bio-Oil injection rate	Bio-Oil injection rate	Always
Gas detection	Gas detection for elevated CO_2, CH_4, or VOC levels in headspace with resolution of at least 0.01%vol	Always

PARAMETER	PARAMETER DESCRIPTION	REQUIRED
Gas isotope stability	Stable isotope compositions of carbon containing species - usually measure by IRMS	Not required but helpful
Corrosion monitoring	Corrosion monitoring	Always
Pressure fall off test	Pressure fall off test	Always
Seismic monitoring	Seismic monitoring	Under certain conditions: as per UIC permit requirements
Microbial assay	Documentation of microbial communities for example by 16S rRNA gene sequencing	Not required but helpful
Bio-oil solidification	Obtain cores of the reservoir to confirm bio-oil solidification	Always
Bio-oil degradation	where possible, in-situ bio-oil plume sample collection - analysis for degradation to VOC	Always
Vertical migration	Utilize tracer studies to demonstrate lack of vertical migration	Always

PARAMETER	PARAMETER DESCRIPTION	REQUIRED
Well integrity	Demonstration of external mechanical integrity - annually	Always
Downhole sonar and seismic measurements	Downhole sonar and seismic measurements	Always
Dissipation interval	Characterize additional dissipation interval below the storage complex to limit downward overpressure propagation	Always

6.1

Biogeochemical Tracer Requirements

There are a variety of biogeochemical tracers that are used widely across the natural sciences and industry for subsurface monitoring and verification. The table below summarizes some of these techniques and is adapted from an earlier version in Tyne et al. 2023[11]. The applicability and usefulness of a particular tracer will depend on the project specification, e.g. δ^{13}C of CO_2 is particularly useful for tracking CO_2 dissolution and mineralization reactions, whereas δ^{13}C of CH_4 is informative about the source of methane and microbial activity.

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	WHAT CAN IT TELL US ABOUT?	LIMITATIONS	APP
δ^{13}C of CO_2	Carbon isotope ratios of CO_2	Not required but helpful	Origin of CO_2 and chemical/biological/physical processes and reactions (e.g. CO_2 mineralization).	Large overlap from difference sources of CO_2 and reactions/processes, which can result in dilution or overprinting of shifts and changes.	Dur Mo Der
δ^{13}C of CH_4	Carbon isotope ratios of methane	Not required but helpful	Origin of CH_4 and chemical/biological/physical processes and reactions (e.g. methanogenesis).	Large overlap from difference sources of CH_4 and reactions/processes, which can result in dilution or overprinting of shifts and changes.	Dur Mo Der
δC of CH_4	Deterium isotope ratios of methane	Not required but helpful	Chemical/biological/physical processes and reactions (e.g. methanogenesis pathway).	Large overlap from difference sources of CH_4 and reactions/processes, which can result in dilution or	Dur Mo Der

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	WHAT CAN IT TELL US ABOUT?	LIMITATIONS	APP
				overprinting of shifts and changes.	
δ^{18}O of $H_2$2O	Oxygen isotope ratios of water	Yes	Origin of water and chemical/biological/physical processes and reactions (e.g. water-rock interaction).	Large overlap from difference sources of water and reactions/processes, which can result in dilution or overprinting of shifts and changes.	Dur Mo Der
δD of H2O	Deterium isotope ratios of water	Yes	Origin of water and chemical/biological/physical processes and reactions (e.g. H2S exchange).	Large overlap from difference sources of water and reactions/processes, which can result in dilution or overprinting of shifts and changes.	Dur Mo Der

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	WHAT CAN IT TELL US ABOUT?	LIMITATIONS	APP
Noble gas concentrations	Concentrations of He, Ne, Ar, Kr, Xe, Ra	Not required but helpful	Sources of gases and fluids, and subsurface reactions/processes. Particularly useful as an inert tracer accompanying injected CO_2	Challenging if injectate and reservoir have similar compositions and very sensitive to contamination.	Dur Mo Der
CO_2/3He	Ratio of CO_2 to 3He	Not required but helpful	Quantify changes in CO_2 in the system, particularly when combined with other tracers such as δ13CCO_2 (e.g. quantify CO_2 dissolution and mineralization)	Need to know starting/unaltered compositions. Challenging if injectate and reservoir have similar compositions and very sensitive to contamination.	Dur Mo Der
Δ47 of CO_2	Clumped isotopes of CO_2	Not required but helpful	Temperature of CO_2 reactions (e.g. carbonate mineralization temperature) - proxy for previous water temperature	Impacts of mixing effects overlap and dilute one and other. Subject to disequilibrium effects which result	Dur Mo Der

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	WHAT CAN IT TELL US ABOUT?	LIMITATIONS	APP
				in inaccurate temperature reconstruction.	
Δ12CH2D2/ Δ13CH3D of CH_4	Clumped isotopes of methane	Not required but helpful	Origin of methane & biological processes	Impacts of mixing effects overlap and dilute one and other.	Dur Mo Der

7.0

Relevant Works

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2. Bio-oil Sequestration as a Viable CDR Pathway, Charm

Industrial, 2023 ↩ ↩ 2 ↩ 3 ↩ 4 ↩ 5 ↩ 6 ↩ 7

3. https://charmindustrial.com/Bio-oil_Sequestration__Protocol_for_Measurement_Reporting_and_Verification.pdf ↩
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5. Ibid. ↩
6. https://www.sciencedirect.com/science/article/abs/pii/S0165993620300868; https://www.sciencedirect.com/science/article/abs/pii/S0165237011001823 ↩
7. https://onlinelibrary.wiley.com/doi/full/10.1111/gcbb.12553 ↩
8. https://charmindustrial.com/Bio-oil_Sequestration__Protocol_for_Measurement_Reporting_and_Verification.pdf ↩
9. This module addresses minimum requirements for injection well design, construction, operation, and monitoring. Although typically these requirements will be addressed in full in the UIC Injection Well permit, the injection of bio-oil is a novel approach for which UIC permitting decisions and requirements are still in development and may not be consistent based on the evaluation and development of permitting approaches by each responsible authority. Therefore, critical concepts and minimum requirements are documented here for consistency and to ensure proper storage reservoir design construction and monitoring to ensure durability of storage. ↩
10. Note that other well classes may be utilized, such as Class II wells, if site specific UIC well permits identify bio-oil as an acceptable injectant. However, Class II wells may not be utilized if the wells are also used for enhanced hydrocarbon recovery (EHR or EHR+) activities. ↩

11. For Class V wells, the well must be permitted and not 'authorized by rule', and must consider the specific emplacement and durable storage of bio-oil in the geologic reservoir. As of writing, the utilization of Class V wells should be limited to wells operating under the Other / Experimental category of Class V wells or other appropriate well type as approved by the UIC permitting authority. ↩ ↩ [2]
12. Area of Review (AOR) means the area surrounding an injection well described according to the criteria set forth in § 40 CFR.146.6, which, in some cases, such as Class II wells, the project area plus a circumscribing area the width of which is either ¼ of a mile or a number calculated according to the criteria set forth in § 146.06. ↩ ↩ [2]
13. See method for Accelerated Aging of Fast Pyrolysis Bio-oil: https://www.nrel.gov/docs/fy16osti/65889.pdf ↩
14. https://www.nrel.gov/docs/fy22osti/82631.pdf ↩
15. Note that detection of organic compounds may not indicate bio-oil migration, particularly when formations are former oil and gas reserves, as residual organic compounds may be present and can be released due to activity in the formation (such as bio-oil injection). In these cases, the project proponent shall investigate such releases and provide determination whether the likely source is from bio-oil or residual hydrocarbons. ↩
16. Cal. Code Regs., tit. 14, § 1724.14, "Pre-Rulemaking Discussion Draft 04-26-17 Updated Underground Injection Control Regulations," (2017). ↩
17. Even if bio-oil solidification cannot be determined from acquired monitoring data, the legal agreement established with the landowner to conduct bio-oil sequestration will ensure no offset activity (i.e., injection or production) will jeopardize sequestered bio-oil for all perpetuity ↩

18. https://www.nrel.gov/docs/fy00osti/27613.pdf ↩
19. https://bioresourcesbioprocessing.springeropen.com/articles/10.1186/s40643-023-00654-3 ↩

Contributors

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Changelog

v1.1

CERTIFIED

Key requirement changes include updates to the Well Construction section to ensure the integrity of pre-existing wells.

v1.0

CERTIFIED

Initial Release.

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